results in the formation of more than one annual ring. Such rings are called **pseudo-** or **false- annual rings**.

Each annual ring corresponds to one year's growth and on the basis of these rings, the age of a particular plant can easily be calculated. The determination of the age of a tree by counting the annual rings is called **dendrochronology**.

Importance of Studying Growth Rings

- Age of wood can be calculated.
- The quality of timber can be ascertained.
- Radio-Carbon dating can be verified.
- Past climate and archaeological dating can be made.
- Provides evidence in forensic investigation.

Dendroclimatology

It is a branch of dendrochronology concerned with constructing records of past climates and climatic events by analysis of tree growth characteristics, especially growth rings.

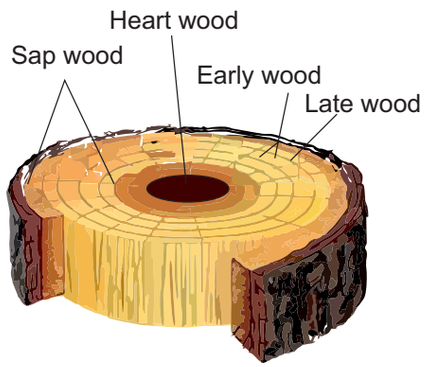


Figure 10.7: Structure of wood – Image shows early wood and late wood



The age of American, *Sequoiadendron* tree is about 3500 years.



Differences Between Spring Wood and Autumn Wood

Spring wood or Early wood	Autumn wood or Late wood
The activity of cambium is faster.	Activity of cambium is slower
Produces large number of xylem elements.	Produces a fewer xylem elements.
Xylem vessels/ trachieds have wider lumen.	Xylem vessels/ trachieds have narrow lumen.
Wood is lighter in colour and has lower density	Wood is darker in colour and has a higher density.

Another feature of wood related to seasonal changes is the diffuse porous and ring porous condition. On the basis of diameter of xylem vessels, two main types of angiosperm woods are recognized.

❖ **Diffuse porous woods**

Diffuse porous woods are woods in which the vessels or pores are rather uniform in size and distribution throughout an annual ring.

Example: *Acer*

❖ **Ring porous woods**

The pores of the early wood are distinctly larger than those of the late wood. Thus rings of wide and narrow vessels occur.

Example: *Quercus*

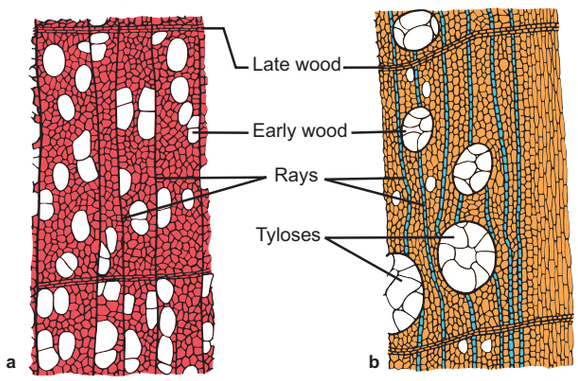


Figure 10.8: Transverse section of wood showing. a. Diffuse porous. b. Ring porous

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